```
[root@abc ~]# nmcli con show ens10 | grep -i dns
ipv4.dns:                               192.168.123.1
ipv4.dns-search:                        example.com
ipv4.ignore-auto-dns:                   no
ipv6.dns:
ipv6.dns-search:
ipv6.ignore-auto-dns:                   no
IP4.DNS[1]:                             192.168.123.1
[root@abc ~]# nmcli conn mod ens10  ipv4.dns '192.168.123.5'
[root@abc ~]# nmcli conn show ens10 | grep dns
ipv4.dns:                               192.168.123.5
ipv4.dns-search:                        example.com
ipv4.ignore-auto-dns:                   no
ipv6.dns:
ipv6.dns-search:
ipv6.ignore-auto-dns:                   no
[root@abc ~]#
```

Figure 9: Changing the DNS using the *nmcli* commands

```
[root@abc ~]# nmcli con show ens10 | grep -i dns
ipv4.dns:                               192.168.123.1
ipv4.dns-search:                        example.com
ipv4.ignore-auto-dns:                   no
ipv6.dns:
ipv6.dns-search:
ipv6.ignore-auto-dns:                   no
IP4.DNS[1]:                             192.168.123.1
[root@abc ~]# nmcli conn mod ens10  ipv4.dns '192.168.123.5'
[root@abc ~]# nmcli conn show ens10 | grep dns
ipv4.dns:                               192.168.123.5
ipv4.dns-search:                        example.com
ipv4.ignore-auto-dns:                   no
ipv6.dns:
ipv6.dns-search:
ipv6.ignore-auto-dns:                   no
[root@abc ~]#
```

Figure 10: Changing the DNS entry in *resolv.conf*

```
[root@abc ~]# firewall-cmd --list-all-zones
block
  interfaces:
  sources:
  services:
  ports:
  masquerade: no
  forward-ports:
  icmp-blocks:
  rich rules:

dmz
  interfaces:
  sources:
  services: ssh
  ports:
  masquerade: no
  forward-ports:
  icmp-blocks:
  rich rules:

drop
  interfaces:
  sources:
  services:
  ports:
  masquerade: no
  forward-ports:
  icmp-blocks:
  rich rules:

external
  interfaces:
  sources:
  services: ssh
  ports:
  masquerade: yes
  forward-ports:
  icmp-blocks:
  rich rules:

home
  interfaces:
```

Figure 11: A complete overview of the firewall configuration

```
[root@abc ~]# iptables -L -t filter -n
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     udp  --  0.0.0.0/0            0.0.0.0/0            udp dpt:53
ACCEPT     tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:53
ACCEPT     udp  --  0.0.0.0/0            0.0.0.0/0            udp dpt:67
ACCEPT     tcp  --  0.0.0.0/0            0.0.0.0/0            tcp dpt:67
ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0            ctstate RELATED,ESTABLISHED
ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0
INPUT_direct  all  --  0.0.0.0/0            0.0.0.0/0
INPUT_ZONES_SOURCE  all  --  0.0.0.0/0            0.0.0.0/0
INPUT_ZONES  all  --  0.0.0.0/0            0.0.0.0/0
ACCEPT     icmp --  0.0.0.0/0            0.0.0.0/0
REJECT     all  --  0.0.0.0/0            0.0.0.0/0            reject-with icmp-host-prohibited

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination
ACCEPT     all  --  0.0.0.0/0            192.168.122.0/24     ctstate RELATED,ESTABLISHED
ACCEPT     all  --  192.168.122.0/24     0.0.0.0/0
ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0
REJECT     all  --  0.0.0.0/0            0.0.0.0/0            reject-with icmp-port-unreachable
REJECT     all  --  0.0.0.0/0            0.0.0.0/0            reject-with icmp-port-unreachable
ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0            ctstate RELATED,ESTABLISHED
ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0
FORWARD_direct  all  --  0.0.0.0/0            0.0.0.0/0
FORWARD_IN_ZONES_SOURCE  all  --  0.0.0.0/0            0.0.0.0/0
FORWARD_IN_ZONES  all  --  0.0.0.0/0            0.0.0.0/0
FORWARD_OUT_ZONES_SOURCE  all  --  0.0.0.0/0            0.0.0.0/0
FORWARD_OUT_ZONES  all  --  0.0.0.0/0            0.0.0.0/0
ACCEPT     icmp --  0.0.0.0/0            0.0.0.0/0
REJECT     all  --  0.0.0.0/0            0.0.0.0/0            reject-with icmp-host-prohibited

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     udp  --  0.0.0.0/0            0.0.0.0/0            udp dpt:68
OUTPUT_direct  all  --  0.0.0.0/0            0.0.0.0/0

Chain FORWARD_IN_ZONES (1 references)
target     prot opt source               destination
FWDI_public  all  --  0.0.0.0/0            0.0.0.0/0           [goto]
FWDI_public  all  --  0.0.0.0/0            0.0.0.0/0           [goto]
```

Figure 12: Firewall information from the iptables command

Although the configuration change has been made to the network manager, */etc/resolv.conf* still has its original configuration. The *nmcli* command must be used to bounce the network connection to apply the configuration changes.

Troubleshooting firewall issues

Once basic network connectivity has been confirmed, a network service may be unavailable because of firewall problems. The firewall rules on the server hosting the network service may have blocked the ports used to access the service. Less frequently, firewall issues can sometimes be a problem on the client machines.

The *firewall-cmd* command can be used to display and adjust current firewall settings. Invoking the command with the *--list-all-zones* option gives a complete overview of the *firewalld* configuration (see Figure 11).

Using the *--permanent* option with *firewall-cmd* displays the persistent settings. Comparing active *firewalld* settings with persistent settings can sometimes identify potential problems.

A quick way to convert runtime rules into permanent rules is with the following command:

```
# firewall-cmd --runtime-to-permanent
```

Firewalld is built on top of net filter functionality. This means iptables can be used to get firewall information at a lower level than *firewall-cmd*.

Similarly, the ip6tables command displays and manages the IPv6 firewall tables. END

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